

Mathematics for Machine Learning

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Positive Definite Matrices

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Definition

A matrix $A \in \mathbb{R}^{n \times n}$ is called **positive definite**, if

$$\mathbf{x}^T A \mathbf{x} > 0, \quad \mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq \mathbf{0}$$

The matrix A is called *positive semi-definite*, if

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Example

Which of the following matrices is positive definite?

$$A = \begin{bmatrix} 4 & -4 \\ -4 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & -4 \\ -4 & 2 \end{bmatrix}$$

Positive Definite Matrices

Proposition

Let V be an n -dimensional vector space with an inner product $\langle \cdot, \cdot \rangle$ and a basis $B = (\mathbf{b}_1, \dots, \mathbf{b}_n)$. Prove that

$$\langle \mathbf{x}, \mathbf{y} \rangle = [\mathbf{x}]_B^T A [\mathbf{y}]_B,$$

where $A_{ij} = \langle \mathbf{b}_i, \mathbf{b}_j \rangle$. Conclude that the matrix A is positive definite.

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Theorem

For a real-valued, finite-dimensional vector space V and a basis B of V it holds that $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ is an inner product if and only if there exists a symmetric, positive definite matrix $A \in \mathbb{R}^{n \times n}$ with

$$\langle \mathbf{x}, \mathbf{y} \rangle = [\mathbf{x}]_B^T A [\mathbf{y}]_B,$$

Proposition

A symmetric matrix A is positive definite if and only if the determinants associated with all upper-left submatrices of A are positive.

Distance and metric

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Any inner product induces a norm

$$\|x\| := \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}.$$

However, not every norm is induced by an inner product (e.g. Manhattan norm or ℓ_1 norm).

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Example

Compute the norm of the vector $[2 \ 2]^T$ w.r.t. the following inner products:

a) $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y}$

b) $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix} \mathbf{y}$

Definition

A function $d : V \times V \rightarrow \mathbb{R}$ is metric, if

- d is positive definite, i.e., $d(\mathbf{x}, \mathbf{y}) \geq 0$ for all $\mathbf{x}, \mathbf{y} \in V$ and $d(\mathbf{x}, \mathbf{y}) = 0 \Leftrightarrow \mathbf{x} = \mathbf{y}$,
- d is symmetric, i.e., $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$ for all $\mathbf{x}, \mathbf{y} \in V$,
- Triangular inequality: $d(\mathbf{x}, \mathbf{z}) \leq d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z})$.

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Proposition

Every norm (inner product) induces a metric:

$$d(\mathbf{x}, \mathbf{y}) := \|\mathbf{x} - \mathbf{y}\| = \sqrt{\langle \mathbf{x} - \mathbf{y}, \mathbf{x} - \mathbf{y} \rangle}$$

Angles and Orthogonality

By Cauchy-Schwarz inequality

$$-1 \leq \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \|\mathbf{y}\|} \leq 1.$$

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Therefore, there exists a unique $w \in [0, \pi]$ with

$$\cos w = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \|\mathbf{y}\|}.$$

The number w is the **angle** between the vectors $\mathbf{x}$ and $\mathbf{y}$.

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Find the angle between $\mathbf{x}$ and $\pm \mathbf{x}$.

Example

Find the angle between $\mathbf{x} = [1, 1]^T$ and $\mathbf{y} = [1, -1]^T$ w.r.t. the inner products given by

a) $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y}$

b) $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \mathbf{y}$

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Definition

Two vectors $\mathbf{x}$ and $\mathbf{y}$ are **orthogonal** if and only if $\langle \mathbf{x}, \mathbf{y} \rangle = 0$, and we write $x \perp y$. If additionally $\|\mathbf{x}\| = 1 = \|\mathbf{y}\|$, i.e., the vectors are unit vectors, then $\mathbf{x}$ and $\mathbf{y}$ are **orthonormal**.

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Remark

Geometrically, we can think of orthogonal vectors as having a right angle with respect to a specific inner product.

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A square matrix $A \in \mathbb{R}^{n \times n}$ is an orthogonal matrix if its columns are orthonormal, i.e. $A^T A = I$.

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Note that for orthonormal matrix $AA^T = I$ and $A^{-1} = A^T$.

Theorem

Suppose S is an orthogonal set of nonzero vectors. Then S is linearly independent.

The Pythagorean Theorem

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The vectors $u, v \in \mathbb{R}^n$ are orthogonal if and only if

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2.$$

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Theorem

Suppose $\{u_1, \dots, u_n\}$ is a set of orthogonal vectors, then

$$\|u_1 + \dots + u_n\|^2 = \|u_1\|^2 + \dots + \|u_n\|^2.$$

Angles and Orthogonality

Proposition

Transformations with orthogonal matrix of transformation preserve the length and the angle w.r.t. to the dot product, i.e.

$$\|A\mathbf{x}\| = \|\mathbf{x}\| \text{ and } w_{\mathbf{x},\mathbf{y}} = w_{A\mathbf{x},A\mathbf{y}}$$

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Definition

Consider an n -dimensional vector space V and a basis $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ of V . The basis B is called an **orthogonal basis**, if

$$\langle \mathbf{b}_i, \mathbf{b}_j \rangle = 0, \quad i \neq j.$$

If additionally the vectors $\mathbf{b}_i$ are unit, then the basis B is called an **orthonormal basis (ONB)**.

Orthonormal Basis

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Example

The canonical/standard basis for a Euclidean vector space $\mathbb{R}^n$ is an orthonormal basis, where the inner product is the dot product of vectors.

Example

Check that the vectors

$$\mathbf{b}_1 = \frac{1}{2} \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix}, \quad \mathbf{b}_2 = \frac{1}{2} \begin{bmatrix} -\sqrt{3} \\ 1 \end{bmatrix}$$

form an orthonormal basis for $\mathbb{R}^2$.

Projection operators

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Remark

The transformation matrix P_π of a projection π is called projection matrix and satisfies $P_\pi^2 = P_\pi$.

Orthogonal Projection onto 1-Dimensional Subspaces

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- $\|\pi_U(\mathbf{x})\| = |\cos w| \|\mathbf{x}\|$, where w is the angle between $\mathbf{x}$ and $\mathbf{b}$.
- With the dot product as inner product the projection matrix P_π satisfying the condition $\pi_U(\mathbf{x}) = P_\pi \mathbf{x}$ is given by

$$P_\pi = \frac{\mathbf{b}\mathbf{b}^T}{\|\mathbf{b}\|^2} = \frac{\mathbf{b}\mathbf{b}^T}{\mathbf{b}^T \mathbf{b}}.$$

Clearly, $\mathbf{b}\mathbf{b}^T$ is a symmetric matrix.

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- $\pi_U(\mathbf{x}) = \sum_{i=1}^m \lambda_i \mathbf{b}_i = B\lambda$, where
 $B = [\mathbf{b}_1, \dots, \mathbf{b}_m] \in \mathbb{R}^{n \times m}$, $\lambda = [\lambda_1, \dots, \lambda_m]^T \in \mathbb{R}^m$.

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- The condition $\mathbf{x} - \pi_U(\mathbf{x}) \perp U$ is equivalent to $\langle \mathbf{b}_i, \mathbf{x} - \pi_U(\mathbf{x}) \rangle = \mathbf{b}_i^T (\mathbf{x} - \pi_U(\mathbf{x})) = 0$ for $i = 1, \dots, m$. Hence

$$\begin{bmatrix} \mathbf{b}_1^T \\ \vdots \\ \mathbf{b}_m^T \end{bmatrix} [\mathbf{x} - B\lambda] = \mathbf{0} \Leftrightarrow B^T(\mathbf{x} - B\lambda) = \mathbf{0} \Leftrightarrow B^T B\lambda = B^T \mathbf{x}.$$

Therefore

$$\lambda = (B^T B)^{-1} B^T \mathbf{x}.$$

The matrix $(B^T B)^{-1} B^T$ is also called **pseudo-inverse** of B , which can be computed for non-square matrices B .

Orthogonal Projection onto General Subspaces

- By $\pi_U(\mathbf{x}) = B\lambda$ we have the projection $\pi_U(\mathbf{x})$ is computed by

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Remark

If the basis $(\mathbf{b}_1, \dots, \mathbf{b}_m)$ is an ONB, i.e. $\mathbf{b}_1, \dots, \mathbf{b}_m$ are unit and orthogonal vectors, then the projection equation simplifies greatly to

$$\pi_U(\mathbf{x}) = BB^T \mathbf{x}.$$

Orthogonal Projection onto General Subspaces

Projections allow us to look at situations where we have a linear system $A\mathbf{x} = \mathbf{b}$ without a solution. Recall that this means that $\mathbf{b} \notin \text{col}(A)$. Given that the linear equation cannot be solved exactly, we can find an **approximate solution**.

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The idea is to find the vector in $\text{col}(A)$ that is closest to $\mathbf{b}$, i.e., we compute the orthogonal projection of $\mathbf{b}$ onto the subspace spanned by the columns of A . This problem arises often in practice, and the solution is called the **least squares solution**.

Definition

The trace of a square matrix $A \in \mathbb{R}^{n \times n}$ is given by

$$\text{tr}(A) := \sum_{i=1}^n a_{ii}$$

in other words, the trace is the sum of the diagonal elements of A .

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Remark

For $A, B \in \mathbb{R}^{n \times n}$ the trace satisfies the following properties:

1. $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$
2. $\text{tr}(\alpha A) = \alpha \text{tr}(A)$
3. $\text{tr}(I_n) = n$
4. $\text{tr}(AB) = \text{tr}(BA)$

Corollary

- In particular, for two vectors $x, y \in \mathbb{R}^n$

$$\text{tr}(\mathbf{x}\mathbf{y}^T) = \text{tr}(\mathbf{y}^T\mathbf{x}) = \mathbf{y}^T\mathbf{x} \in \mathbb{R}.$$

- If $A \sim B$ (i.e. $\exists Q$ s.t. $A = QBQ^{-1}$), then $\text{tr}A = \text{tr}B$
- All transformation matrices A_T of a linear mapping $T : V \rightarrow V$ have the same trace.

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Definition

For $\lambda \in \mathbb{R}$ and a square matrix $A \in \mathbb{R}^{n \times n}$

$$p_A(\lambda) = \det(A - \lambda I) = c_0 + c_1\lambda + c_2\lambda^2 + \dots + c_{n-1}\lambda^{n-1} + (-1)^n\lambda^n$$

$c_0, \dots, c_{n-1} \in \mathbb{R}$, is the **characteristic polynomial** of A . In particular,

$$c_0 = \det(A), \quad c_{n-1} = (-1)^{n-1}\text{tr}(A).$$